

AB123C: Distortion-Entangled Bodies in the AI-gorithmic Age

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Figure 1: The robotic art installation "AB_{123C}".

Abstract

This paper presents the robotic installation AB_{123C}, an information evolution system composed of five interacting robotic agents. Rather than preserving informational fidelity, the system stages distortion as a structural principle—a generative condition through which information acquires form. As descriptive content circulates across a chain of machines, it undergoes semantic drift, structural fragmentation, and emergent recomposition. These transformations do not reflect malfunction, but express the entangled ontology of mediated systems. By integrating algorithmic indeterminacy, material reconfiguration, and environmental perturbation, the installation visualizes how meaning arises through iterative misalignment. Within this machinic ecology, truth is not preserved—it is enacted,

shaped through continuous entanglement, differential deferral, and distributed agency.

CCS Concepts

• Applied computing → Media arts.

Keywords

Robot art, Information transmission, Installation design

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1 Introduction

In contemporary informational ecologies, messages no longer follow stable trajectories. Instead, they mutate—semantically, structurally, and contextually—across algorithmic and material infrastructures. With each relay, meaning shifts, fragments, and recombines, exposing the fragile and contingent nature of communication [Carmi 2020; Purwanti and Suteja 2024]. What emerges is not informational clarity, but a space of ongoing transformation—where distortion becomes the method of inscription, and entanglement the grammar of flow.

The installation AB₁₂₃C explores these conditions through a chain of robotic agents. Beginning with Robot A, which performs visual analysis and generates textual descriptions from physical objects, information is then successively modulated by three intermediate units (B₁₂₃), each applying distinct semantic or structural mutations. Robot C completes the sequence by translating the accumulated transformations into a visual-material output. This process does not aim to reconstruct the original message, but to render visible the trajectories of its becoming.

Crucially, AB₁₂₃C is not a closed relay but a responsive field. The modularity of the B-units allows for rearrangement, substitution, and interference—exposing the transmission to both programmed variation and environmental noise. These perturbations are not anomalies; they are constitutive agents within the system. The installation thus materializes informational drift as an entangled choreography, where content evolves not in spite of distortion, but through it.

Through this lens, truth is not a stable referent but a provisional pattern—emerging at the intersection of machine logic, sensory registration, and interpretive delay. AB₁₂₃C does not seek to fix meaning, but to trace how it splinters, stretches, and reforms across dynamic and interdependent systems.

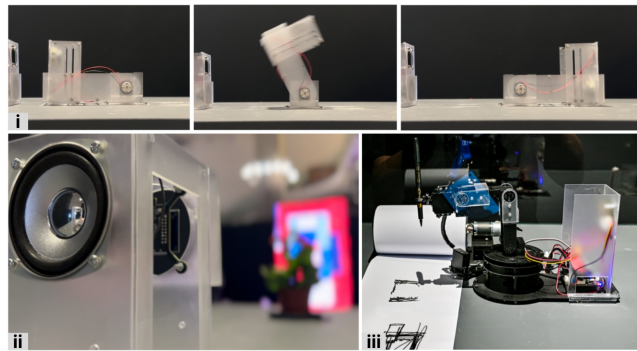


Figure 2: (i) The mechanical reversal structure of Robot B (ii) The image-processing unit, Robot A (iii) The robotic arm of Robot C performing the drawing output.

2 AB₁₂₃C: Sequence of Robots

2.1 Design Consideration

Naming: The name AB₁₂₃C succinctly and intuitively captures the structural features of the system. A, B, and C represent three

types of robotic units with distinct functions: A serves as the sensor and generator of information, B as the transmitter and processor, and C as the final expresser. The subscript "123" indicates variable B-type nodes in the transmission chain. Inspired by the "Telephone" game, this robotic system transforms a familiar social interaction into a metaphor for information transmission and mutation.

Transmission: B₁₂₃ serve as the primary conduit for information flow. It consists of a rotating mechanical arm and a terminal box (as shown in Figure 2 i). Upon flipping, each B causes the rectangular acrylic box at the end to impact the ground, revealing the other half of the box, thus achieving a natural switch from receiver to speaker. This symbolizes that each node in the information dissemination network acts as a receiver, shaper, and transmitter, illustrating the multiple evolutions and re-creations of information throughout the transmission process (Figure 3). As information is transmitted repeatedly, B₁₂₃ flip like waves, materializing the vector of information flow in physical space.

Drawing: Robot C's mechanical arm performs a live drawing process, sharing space with the audience (Figure 2 ii). This transforms information presentation into a spatial performance, making visible the translation from text to image. The physical act of drawing introduces natural variations through pen contact, mechanical movements, and environmental factors. These "organic errors" represent the inevitable distortions in information transmission, reflecting on information authenticity in our post-truth era.

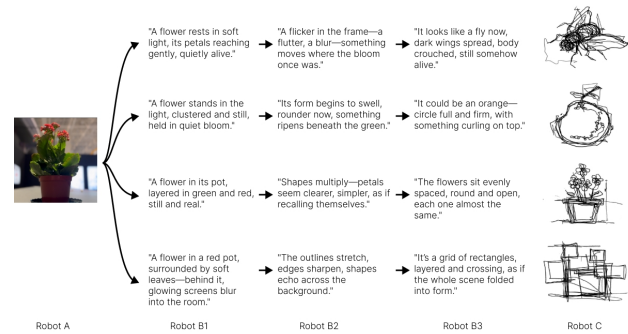


Figure 3: The Transmission Process of Information from the Same Plant

Materials: The robots feature transparent acrylic, exposed mechanics, and visible circuit boards, creating a visually transparent design that reveals their operational processes. Yet this physical transparency contrasts with the opacity of information processing within. As data moves through the system, it undergoes multiple transformations, making its meaning increasingly ambiguous. This paradox reflects a key aspect of our post-truth era: despite transparent technology and open information systems, meaning itself can become obscured through layers of interpretation.

Observed Object: Plants were chosen as the subject for their familiar yet complex nature (Figure 2 iii). They serve as both the initial information source and a metaphor for truth. Through AB₁₂₃C's

process of observation, transmission, and drawing, this "truth" undergoes multiple transformations. The installation explores how information changes from its original form to final representation, prompting reflection on how truth is perceived, communicated, and reconstructed in our post-truth era.

2.2 Dynamic Structural Modulation

In "AB₁₂₃C", audiences can engage with the dynamic flow of information at various levels, directly becoming catalysts for its dissemination and transformation:

- **Change the Observed Object:** Audiences can replace the original subject, such as substituting other objects for the plants, directly altering the starting point of information creation. Each substitution triggers new semantic pathways, illustrating how information is reshaped based on different origins.
- **Disrupt the Transmission Chain:** As information flows through B₁₂₃, audiences can interact through movement, observation, listening, or vocalization. This interaction simulates real-world information flow: transmission is constantly exposed to external interference, with audience intervention revealing both the fragility of information and its complex evolution.
- **Reshape the System Structure:** Audiences can also adjust the structure of the installation by increasing or decreasing the number of B₁₂₃ units or rearranging their sequence. Such physical changes fundamentally alter the network architecture of information transmission, reflecting the decisive impact of structure on meaning formation.

Through multilayered interactions, the installation shows how information changes as it moves through the system. Each audience interaction affects the network's behavior, demonstrating how truth becomes collectively shaped rather than remaining fixed.

2.3 Implementation

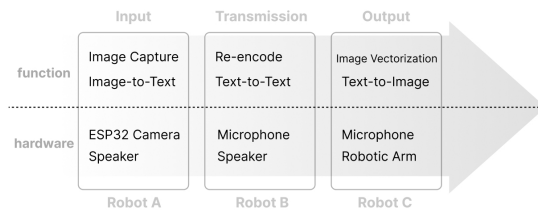


Figure 4: Core Functions and Associated Hardware Modules

Hardware: The installation is built upon ESP32 microcontroller platforms, integrating servo motors and various sensors to drive actuators and handle data collection and transmission among units. Robot A is equipped with a camera module that captures real-time image data and invokes a lightweight local vision model to extract features and generate descriptive statements in both Chinese and English. B₁₂₃ utilizes multiple servo motors to control a multi-joint

robotic arm, executing repetitive flipping actions. Each flip represents a semantic transformation, simulating how information is reshaped—through recombination, exaggeration, or omission—during transmission. Robot C comprises a complete drawing robotic arm system, including an end-effector and a receiver module. Upon receiving the final textual statement from B₁₂₃, it calls the large language model to generate an image corresponding to the input. The generated image is then vectorized using the Potrace algorithm. Inverse kinematics is applied to calculate the joint angles required by the robotic arm. Prior to execution, the vectorized path is further refined using linear or spline-based interpolation to smooth the trajectory and ensure precise, continuous motion. The interpolated path is discretized into frame-wise commands suitable for servo control, and the system drives the servo motors via real-time angle solving and PWM signal output to draw the image line-by-line.

Structure: The terminal flipping structure of B₁₂₃ consists of a rectangular acrylic box, with a speaker mounted on the top and an infrared receiver on the bottom. The box is embedded in two parallel sliding grooves on the robotic arm via two metal rods, allowing it to move vertically. Each time the arm descends and contacts the ground, the inertia and structural design cause the box to lift along the grooves, flipping to reveal the previously hidden section. This motion symbolizes the decoding or reinterpretation of information at the receiving end.

Agents: Each machine unit is driven by an agent powered by a large language model (LLM), performing distinct tasks via prompt engineering. Robot A's agent integrates a vision-language understanding model to transform visual input into structured semantic descriptions. B₁₂₃ houses three stylistically distinct agents: B₁ specializes in summarization, B₂ in exaggeration, and B₃ in abstraction or ambiguity. These simulate different re-narration styles, mimicking the distortions of information through various intermediaries. Robot C's agent receives the final statement, generates a black-and-white image via Zhipu Qingyan, vectorizes it with Potrace, applies interpolation and smoothing to the resulting path, then calculates angle sequences using inverse kinematics to execute the drawing with its robotic arm.

3 Distortion as Form: Entangled Becoming

Truth must achieve self-actualization through mediated processes of distortion[Carmi 2020], a constitutive entanglement materialized as artistic praxis in AB₁₂₃C. The installation transduces Derrida's Différance—wherein meaning emerges through differential deferral—into perceptible trajectories of informational metamorphosis via semantic transposition[Derrida 1982], structural reconfiguration, and volumetric mutation. The deliberate integration of algorithmic stochasticity and spectator intervention transforms Heideggerian "Being-in-the-world" into an onto-praxis of artistic becoming[Bohr 1937], where dissipation and proliferation within human-machine interaction chains manifest as epistemic reconfigurations.

Noise within communication theory undergoes critical reframing: classical fidelity paradigms yield to the apparatus's revelation of nuanced interrelations[Shannon 1948]. Physiochemical noise (anthropic intervention/mechanical resonance) and hermeneutic

noise (heterogeneous robotic processing) interlace through engineered aleatory mechanisms, constituting distortion as a generative engine for informational reconfiguration. The revitalization of original signals through novel connectivities substantiates distortion not as fallacy but as the ontological condition of transmission.

The B₁₂₃ robotic ensemble constitutes an open transmission system whose nodal stochasticity, positional interchangeability, and configurational fluidity mirror the pluridirectional flux of digital-era information traversing social platforms. Each robotic agent operates as an inflection point for semiotic mutation, wherein re-encoding of source material and emergent contextualities render uncertainty an immanent attribute of dissemination. This epistemic shift dismantles the "echo chamber effect" of closed systems, asserting distortion as the essential coefficient of informational becoming[Pariser 2011].

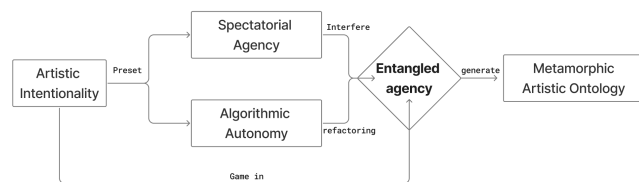


Figure 5: The Dialectics of Control in Open Systems: The Transmutation of Artistic Agency from Direct Manipulation to Systemic Architectonics

Through the agential strategy of "distortion as form," artists forge entangled networks of human/machine/algorithmic actors while relinquishing direct control. Spectatorial agency (as physical noise source) and algorithmic autonomy (as noise generator) are subsumed within a metarule framework (e.g., complementary transmission structures), instantiating a system of controlled indeterminacy[Latour 2012]. The voluntary abandonment of definitive formal control, authoritative interpretative claims, and linear predictability is counterbalanced by the work's evolutionary potential, dialectical cognitive tensions, and the quantum ontology of artistic being.

The visual output at robot C, those pixel arrays deemed "deviant" by classical aesthetics, materializes as ontological scar tissue forged through technological mediation. It inscribes the mutual contamination of observer and observed systems, rendering artistic truth a dynamic negotium emerging from consensual formation. Within semantic drift and structural fragmentation, static entities complete their quantum leap toward entangled existence. Herein lies ontological agency: through polyphonic mediatory negotiations, the work acquires sustained resistance against univocal interpretation, embodying generative vitality through perpetual becoming.

4 Conclusion

AB₁₂₃C is an art installation combining artificial intelligence and robotics, where five collaborative robots form an intelligent information transmission system. Audiences can actively intervene in the process—through sound interference, position changes, or substituting objects—directly influencing AI's perception and creative output. The transparent design and flipping robotic mechanisms

make the flow of information tangible, offering both an artistic exploration of contemporary communication phenomena and a reflective lens on the nature of information. Through its operation, the installation reveals distortion as a fundamental aspect of information, with semantic shifts and structural reorganizations highlighting its multidimensional evolution. Reality, shaped through observation and reinterpretation at each stage, emerges as a dynamic state dependent on context, with post-truth reflecting truth through specific frameworks. Robot C's drawing practice further exposes the tension between creation and deviation, where each misstep opens poetic possibilities, and every act of creation reinterprets the original. This tension defines a key aesthetic dimension of information transmission.

References

- Niels Bohr. 1937. Causality and complementarity. *Philosophy of Science* 4, 3 (1937), 289–298.
- Elinor Carmi. 2020. *Media distortions: Understanding the power behind spam, noise, and other deviant media*. Peter Lang International Academic Publishers.
- Jacques Derrida. 1982. *Margins of philosophy*. U of Chicago P (1982).
- Bruno Latour. 2012. *We have never been modern*. Harvard university press.
- Eli Pariser. 2011. *The filter bubble: What the Internet is hiding from you*. Penguin Press.
- Christina Purwanti and Hanna Suteja. 2024. Language distortion in social media: An interdisciplinary study. In *Inclusive, Sustainable, and Transformational Education in Arts and Literature*. Routledge, 165–169.
- Claude Elwood Shannon. 1948. A mathematical theory of communication. *The Bell system technical journal* 27, 3 (1948), 379–423.